

## 3-Day Work Report

### Overview:

Over the last three days, our team focused on automating the video processing and annotation pipeline. We worked on downloading videos, setting up automation scripts, and improving the annotation workflow to make the process faster and easier.

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### Work Completed:

#### 1. Video Download and API Integration

- Automated video downloads using the **Google Drive API**.
- Created and configured a project in **Google Cloud Console** with API credentials.
- Developed a Python script to automatically download data from the API and save it on the server.
- Handled API rate-limit issues and added retry mechanisms for stable downloading.

#### 2. Environment Setup and Frame Extraction

- Set up the required environment on the server.
- Created a **frame extraction script** to automatically capture frames from downloaded videos.
- Verified that the process runs smoothly without manual steps.

#### 3. Roboflow Automation

- Built an **automation script** to upload extracted frames directly to **Roboflow**.
- This replaces the previous manual upload method, making the workflow fully automated.

#### 4. Workflow Integration

- Combined all processes into a single automated pipeline:  
**Download video → Extract frames → Upload to Roboflow.**
- Successfully tested the complete end-to-end workflow.

## 5. Annotation Work

- Continued annotating additional images to improve dataset quality and coverage.

## 6. Automatic Annotation Research

- Researched ways to **make the annotation process automatic and easier.**
- Currently working on a semi-automated solution — it won't be fully automatic yet, but it will greatly reduce manual effort.

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## Next Steps:

- Monitor and optimize the automated pipeline for large-scale data.
- Continue improving semi-automated annotation tools.
- Maintain annotation quality checks and dataset updates.